Question	Answer	Marks	Partial Marks
2(b)	11	**3**	**M2** for $\frac{220-195.8}{220}[\times 100]$ or for $[100-]\frac{195.8}{220}\times 100$ or **M1** for 220 – 195.8 or for $\frac{195.8}{220}$ or a correct implicit equation for percentage reduction or for $\frac{195.8-220}{220}$
2(c)	84	**3**	**M2** for $\frac{63}{1-\frac{25}{100}}$ oe or **M1** for associating 63 with (100 – 25)% or a correct implicit equation for the original price.
3(a)	662.45	**2**	**M1** for $600\times\left(1+\frac{2}{100}\right)^5$ oe
3(b)(i)	800	**2**	**M1** for $x\left(1+\frac{5}{100}\right)^2=882$ oe or **SC1** for answer 82
3(b)(ii)	5 nfww	**2**	**M1** for trial with $882\times\left(1+\frac{5}{100}\right)^n$ with $n>1$
4(a)(i)	955 or 955.0 to 955.2	**2**	**M1** for $2\times\pi\times 8\times 19$ oe
4(a)(ii)	812 or 811.7 to 811.9...	**2**	**FT** *their* (i) × 0.85 **M1** for *their* (i) × 0.85 or *their* (i) × 85
4(b)(i)	$\frac{\frac{4}{3}\times\pi\times 6^3}{\frac{1}{3}\times\pi\times 8^2}$ or cancelling clearly seen to reach 13.5	**M2**	**M1** for $\frac{4}{3}\times\pi\times 6^3=\frac{1}{3}\times\pi\times 8^2\times h$
4(b)(ii)	15.7 or 15.69...	**2**	**M1** for $8^2+13.5^2$ or better
4(b)(iii)	394 or 395 or 394.3 to 394.6...	**1**	**FT** π × 8 × *their* (b)(ii)

Question	Answer	Marks	Partial Marks
4(c)	567	**3**	**M2** for $\frac{168}{V} = \left(\frac{80}{180}\right)^{\frac{3}{2}}$ oe or better or **M1** for $\left(\frac{180}{80}\right)^{\frac{1}{2}}$ or $\left(\frac{80}{180}\right)^{\frac{1}{2}}$ oe seen or better
4(d)	51.3 or 51.34...	**3**	**M2** for tan = $\frac{5}{4}$ oe or **M1** for recognition of angle *PBX*
5(a)	4.29 or 4.285 to 4.286	**3**	**M2** for $\frac{150}{\frac{450}{3.6} - \frac{120}{4} - \frac{180}{3}}$ or **M1** for [time =] 120 ÷ 4 or 180 ÷ 3 or 450 ÷ 3.6 or $3.6 = \frac{150+180+120}{\text{total time}}$
5(b)	82.8 or 82.81 to 82.82 using cosine rule	**4**	**M2** for $\frac{150^2 + 120^2 - 180^2}{2 \times 150 \times 120}$ or **M1** for $180^2 = 120^2 + 150^2 - 2 \times 120 \times 150 \cos(...)$ **A1** for $\frac{4500}{36000}$ oe
5(c)(i)	127.2 or 127.1 to 127.2 or 127	**1**	**FT** 210 – *their* (b)
5(c)(ii)	307.2 or 307.1 to 307.2 or 307	**2**	**FT** 180 + *their*(c)(i) **M1** for 180 + *their* (c)(i)
5(d)	15 or 14.99 to 15.04	**2**	**M1** for $\cos(\textit{their}\ (\text{b})) = \frac{\text{dist}}{120}$ oe
6(a)(i)	34	**1**	
6(a)(ii)	18	**2**	**B1** for [l.q. =] 25 or [u.q. =] 43 seen
6(a)(iii)	60	**2**	**M1** for 140 written
6(b)(i)	49	**1**	
6(b)(ii)	20	**1**	
6(b)(iii)	10	**1**	
6(b)(iv)	220	**2**	**M1** for 3 × 1 + 1 × 2 + 3 × 5 + 2 × 10 + 4 × 20 + 2 × 50
6(b)(v)	14.7 or 14.66 to 14.67	**1**	**FT** *their* (iv) ÷ 15